Title: On High-dimensional Feature Selection Using Dimensionality Reduction Methods

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Authors: C. Novak, E. Okafor (National Laboratory for Applied Computation)

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Abstract:

High-dimensional feature selection remains a critical challenge for statistical learning, particularly in urban mobility analysis. This paper proposes a framework built on dimensionality reduction methods that explicitly accounts for causal effect estimation from observational data. We evaluate the framework on open census microdata and compare it against 4 baseline methods. Experimental results show that the proposed approach improves forecast MAPE by 9.8% while keeping RMSE within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of statistical learning has changed how researchers and practitioners approach high-dimensional feature selection, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Despite considerable progress, existing solutions for high-dimensional feature selection tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in silhouette coefficient frequently come at the expense of forecast MAPE, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how dimensionality reduction methods interact with gradient-boosted decision trees when applied to high-dimensional feature selection.

Our work is motivated by the growing demand for urban mobility analysis, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of high-dimensional feature selection, a practical design built on dimensionality reduction methods, and an empirical evaluation on open census microdata. Understanding causal effect estimation from observational data is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Iyer et al. (2014) introduced an early framework based on dimensionality reduction methods, reporting a 41.8% improvement in forecast MAPE over previous baselines. Subsequent work by Sharma et al. (2011) extended this line of research to urban mobility analysis, although the evaluation was restricted to small-scale settings. A survey by Rossi et al. (2016) categorised existing approaches to high-dimensional feature selection into three families and highlighted the lack of standard benchmarks. Researchers working on statistical learning have explored gradient-boosted decision trees as an alternative, with Larsen et al. (2016) noting that the approach scales poorly beyond 32 components. Kowalski et al. (2019) analysed causal effect estimation from observational data and showed that purely heuristic solutions cannot guarantee consistent forecast MAPE.

More recently, Silva et al. (2021) combined dimensionality reduction methods with gradient-boosted decision trees, achieving a 10.1% reduction in overhead on open census microdata. Several studies have criticised the reliance on synthetic workloads, arguing that results on open census microdata may not generalise to urban mobility analysis. In contrast to these efforts, our work treats high-dimensional feature selection and causal effect estimation from observational data jointly rather than as independent concerns. Despite this body of

literature, no prior study has systematically compared dimensionality reduction methods and gradient-boosted decision trees under identical experimental conditions.

3. Methodology

We formalise high-dimensional feature selection as a constrained optimisation task in which the objective is to maximise forecast MAPE while keeping AUC-ROC within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on dimensionality reduction methods, and a validation layer that incorporates gradient-boosted decision trees. To ensure reproducibility, every experiment was repeated 10 times with independent random seeds, and we report mean values together with 95% confidence intervals. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies dimensionality reduction methods to reduce the effect of causal effect estimation from observational data, followed by a calibration step that stabilises AUC-ROC. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 4 baseline methods, including a conventional approach built on gradient-boosted decision trees and two recent variants reported in the literature. The proposed design improves forecast MAPE by 9.8% relative to the strongest baseline and reduces forecast MAPE by 24.5%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the gradient-boosted decision trees component reduces forecast MAPE by approximately 12.0%, confirming its contribution. A sensitivity analysis over 1200 parameter settings shows that performance remains stable, with variation in AUC-ROC below 13.8%. Threats to validity include the limited diversity of open census microdata and the possibility that deployment in urban mobility analysis introduces conditions not captured in our setup.

4. Conclusion

This paper investigated high-dimensional feature selection through the lens of dimensionality reduction methods, and demonstrated that a carefully designed pipeline can improve forecast MAPE by 9.8% on open census microdata. The findings suggest that statistical learning will continue to benefit from closer integration of dimensionality reduction methods and gradient-boosted decision trees, particularly in urban mobility analysis. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of causal effect estimation from observational data. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for high-dimensional feature selection.